Beyond aggregating a building index: spatially honest pluvial flood learning and adaptive H3 screening with an explicit rainfall-conditioned cell index

Working manuscript (methods / IJDRR-shaped)

Status: Matured draft constrained to live Lower Manhattan open-data pilot metrics. Incomplete items marked 待补充.

Public code and documentation: https://github.com/Coucou2016/pluvial-flood-risk-DGGS-H3

Terminology ledger

Term	Canonical meaning
H3	Uber hexagonal Discrete Global Grid System
Open labels	DEP stormwater polygons, 311 flooding points, USGS Ida HWM — not PFIb
Spatial H3-block CV	GroupKFold holding out coarse H3 parent blocks
`PFI_h(c,r)`	Model rainfall-conditioned flood probability/index for cell `c` under rainfall condition `r`
PFIb	7Analytics / Svellingen building-level pluvial flood index — **not used here**
LM pilot	Lower Manhattan open-data pilot (`n_cells=141`)
Expanded pilot	`manhattan_expanded` open-data pilot (`n_cells=956`), still not citywide

Abstract

Urban pluvial flood screening increasingly relies on machine learning and multi-resolution grids, yet many operational indices remain tied to proprietary insurance labels and evaluation protocols that ignore spatial leakage. Building on the H3 DGGS framing popularised for scalable pluvial mapping, we present an **open-label** hexagonal **learning protocol** for Manhattan pilots: multi-source public flood indicators are joined to H3 cells, gradient-boosting models are assessed with **H3-block spatial cross-validation**, scale-loss is quantified with a Jaccard/F1 hotspot ladder, and an **adaptive** refinement stage screens parents using trained cell scores. We define `PFI_h(c,r)` explicitly as a rainfall-conditioned hex flood probability/index — a model output, not feature importance and not the proprietary PFIb product. On the live open-data pilot table (`n=141` cells; open-data assembly), spatial CV accuracy was **0.784 ± 0.069** (5 folds) with mean F1 **0.866**, but the positive class

dominates (80% of held-out cells), and a trivial always-positive majority baseline reaches **0.808** accuracy and **0.893** F1 — so these blocked metrics are **not** reported as classification skill; continuous-risk R^2 under the same folds was near zero (**0.030 ± 0.343**) and is likewise not claimed as skill. A second, larger open-data pilot (`manhattan_expanded`; `n=956` cells, 28 blocks, 47.9% positive) is more balanced: spatial CV accuracy **0.642 ± 0.148** exceeds the always-positive majority baseline (**0.479**), and continuous-risk R^2 is **0.525 ± 0.112**, while mean F1 (**0.608**) remains below the always-positive F1 (**0.648**) — so classification discrimination is still not claimed, but the model shows above-majority accuracy and non-trivial continuous signal at this larger scale. Adaptive refinement used about **57%** as many cells as a uniform R11 grid while refining high-score parents — a **cell-count** comparison only. We do **not** claim citywide skill, PFIb reproduction, radar rainfall, or rainfall-conditioned discrimination: event rainfall remains a synthetic constant hook, and scenario tables show **no within-cell `PFI_h` change across rainfall intensities** (待补充). Synthetic demonstrations are excluded from scientific evidence.

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1 Introduction

Intense short-duration rainfall can overwhelm urban drainage and generate pluvial flooding with limited warning. City-scale assessment needs representations that are (i) computationally scalable, (ii) updateable when new observations arrive, and (iii) evaluable without silent spatial leakage. Discrete Global Grid Systems (DGGS), and in particular Uber H3, provide nested hexagonal cells that support multi-resolution aggregation and neighbour queries.

Svellingen et al. (2026) demonstrated that machine-learning-derived building-level pluvial susceptibility (PFIb) can be aggregated into H3 cells to reduce spatial query cost by roughly two orders of magnitude, while also showing that hotspot sets can diverge sharply across resolutions (Jaccard similarity ≈ 0.14 between fine street-level and coarser neighbourhood hotspots on their proprietary stack). That work motivates H3 as a communication and screening fabric, but its focus is the aggregation and communication of a proprietary index, not an open, spatially honest learning protocol for jurisdictions that cannot access insurance claims.

This manuscript therefore asks: **can an open multi-source label stack on H3 support spatially blocked evaluation, diagnose scale-loss, and drive adaptive refinement with an explicit rainfall-conditioned cell index — without claiming PFIb reproduction?** We contribute a **methods protocol**, not a citywide operational product:

1. An open-label H3 feature/label assembly protocol with explicit provenance fields for assembly mode, feature source, label source, and rainfall source.
2. Primary **blocked evaluation** reporting via spatial H3-block CV, with random splits demoted to diagnostics and class-prevalence baselines disclosed alongside accuracy/F1.
3. A scale-loss Jaccard/F1 ladder and an adaptive H3 ablation screened by trained `PFI_h`, plus a binding definition of `PFI_h(c,r)` that is neither feature importance nor PFIb.

Evidence is restricted to a Lower Manhattan open-data pilot (`n=141`). Claims of citywide skill, radar rainfall, PFIb parity, or rainfall-conditioned discrimination are out of scope until those gaps close (待补充).

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2 Related work

From building indices to hexagonal screening. Escalating short-duration extremes have pushed operational risk assessment toward leaner, updateable representations. Svellingen et al. (2026, IJDRR) show that a machine-learning building-level pluvial susceptibility index (PFIb) can be aggregated into H3 cells, cutting geometry-heavy spatial query cost by roughly two orders of magnitude (~98% in their report) while exposing a clear resolution trade-off: fine street-level hotspots are largely invisible at coarser neighbourhood grids (Jaccard ≈ 0.14 between their R13 and R10 hotspot sets). A related SSRN preprint develops the same H3 indexing narrative (Svellingen et al., SSRN). That line of work establishes H3 as a scalable communication fabric for proprietary indices; its stated focus is aggregation and communication rather than open-label learning or spatially blocked evaluation for jurisdictions without insurance claims. We therefore treat PFIb→H3 aggregation as the closest precedent and as an explicit claim boundary (no PFIb reproduction; no numerical equality to their Jaccard). Note also that Svellingen et al. reuse the symbol `PFI_h` for their H3-aggregated PFIb; in this manuscript `PFI_h(c,r)` denotes a separately defined rainfall-conditioned model probability/index and is **not** derived from PFIb.

Hexagonal DGGS as analysis substrate. Independently of insurance labels, Li et al. (2022) use an ISEA3H hexagonal DGGS for multi-scale flood mapping under climate scenarios, emphasising resolution-consistent predictors. Their contribution supports DGGS nesting as a scientific substrate. Our focus differs: an Uber H3 **learning and evaluation protocol** on open urban flood indicators, not climate-scenario inundation mapping alone.

Spatial holdouts in GeoAI. Random train/test splits routinely inflate skill under spatial autocorrelation. Sun, Hu, Lakhanpal & Zhou (2023) establish the rationale for spatially separated validation in geospatial machine learning; this study operationalises that rationale with `GroupKFold` over coarse H3 parent IDs, so that entire parent blocks are withheld and primary reporting aligns with leakage-aware practice rather than treating random accuracy as the headline metric.

Urban pluvial ML susceptibility. City-scale studies map susceptibility from DEM, land cover, and drainage proxies with classical ML (e.g. Bersabe & Jun, 2025, Seoul). The present study combines open multi-source labels, H3 nesting, adaptive refinement screened by trained scores, and an explicit rainfall-conditioned cell index in one reproducible pipeline with blocked CV as the primary evaluation. That combination — and the honesty constraints that follow from a small open-data pilot — is the gap this manuscript addresses.

Honest novelty (one paragraph). Relative to PFIb→H3 aggregation papers, we do not improve or reproduce a proprietary building index; we ask whether public flood indicators can support spatially blocked learning on H3. Relative to hexagonal DGGS flood mapping, we emphasise evaluation protocol and adaptive screening rather than scenario inundation alone. Relative to urban pluvial ML susceptibility maps, we add H3-block CV as the primary **blocked evaluation** report, an open-label scale-loss ladder, adaptive cell-count diagnostics, and a binding non-PFIb definition of `PFI_h(c,r)` — while refusing citywide, radar, and rainfall-discrimination claims until stronger evidence exists.

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3 Study area and data

Extent. Lower Manhattan bounding box (approx. 74.02–73.97°W, 40.70–40.76°N), as documented in the public repository configuration and download manifest. This is a **pilot extent**, not citywide New York City.

Live layers (open downloads, August 2026). DEM (USGS 3DEP subset), DEP stormwater flood polygons, building footprints, USGS Ida high-water marks, 311 flooding points (ArcGIS/CDN mirrors), FEMA Sandy inundation (negative control only), NLCD impervious fraction, NHDPlus HR hydro vectors (`dist_stream_m` as a **distance-to-water** proxy in a tidal/shoreline setting). FloodNet is stub/opt-in. Event rainfall is a **synthetic constant** Ida-like hook, not radar.

Data sources. Elevation is from the USGS 3D Elevation Program (3DEP); impervious fraction from the Multi-Resolution Land Characteristics Consortium National Land Cover Database (NLCD); hydrography (`dist_stream_m`) from NHDPlus High Resolution; flood labels from the New York City Department of Environmental Protection stormwater flood polygons, NYC 311 flooding service requests, and USGS Hurricane Ida high-water marks; the negative control from FEMA Sandy storm-surge inundation. Provider/access citations are listed in References; version and download dates are recorded in the repository download manifest.

Provenance. Training table assembly reports open-data assembly with observed static features; rainfall may still carry an event-raster provenance tag rather than gauge/radar observation.

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4 Methods

4.1 H3 representation

Training uses H3 resolution **R9** (configurable). Scale-loss diagnostics assemble labels at fine resolution **R10** and roll hotspots to parents R9/R8. Adaptive refinement expands selected parents to **R11**. Joins use EPSG:4326.

4.2 Features and open labels

Static predictors: elevation, slope, flow-accumulation proxy, impervious fraction, urban land-cover flag, building density, distance-to-water. Rainfall enters as rainfall intensity for scenario conditioning; observed static columns are tracked separately from rainfall so a constant synthetic grid is not marketed as radar.

Labels combine DEP polygon area fractions with 311 / Ida point counts into flood-risk / flood-class targets as **observed flood labels** (not ground truth). We do not use PFIb. FloodNet joins only when explicitly enabled and a non-empty layer exists.

4.3 Models and baselines

Primary learner: gradient-boosting classifier + continuous risk regressor. Baselines: L2 logistic / linear, and an elevation–impervious–slope–TWI-like ponding rule. In-sample metrics are optimistic references only.

4.4 Spatial H3-block cross-validation

Cells are grouped by coarse H3 parents; GroupKFold holds out entire blocks. Per-fold metrics are archived with the pilot run. Random i.i.d. splits are not primary.

4.5 Scale-loss diagnostics

Fine-resolution hotspot sets (top quantile of risk) are compared to parent-aggregated hotspots via Jaccard and F1 for mean/max/p90 rollups. Numeric values are **not** equated to Svellingen et al.'s PFIb Jaccard ≈ 0.14.

4.6 Adaptive refinement

After training, cell scores (`PFI_h` / flood probability) screen parents for refinement. Metrics: mixed-resolution cell counts versus fixed coarse and versus uniform fine (adaptive/uniform cell-count ratio). The pilot uses trained `PFI_h` as the screening score.

4.7 Event-conditioned `PFI_h(c,r)`

$$
\mathrm{PFI}_h(c,r)=\widehat{P}(Y_c=1\mid X_c,r)
$$

Static features X_c are held fixed while rainfall condition r varies across named scenarios. This is a **model output**, not SHAP/permutation importance and not PFIb.

4.8 Negative control

FEMA Sandy coastal inundation overlaps are reported for separation checks only and are never training labels.

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5 Experiments

ID	Description	Role
E1	Assemble open LM table	Feature/label table for the pilot bbox
E2	Spatial CV	Primary blocked accuracy / F1
E3	Baselines	Diagnostic comparison
E4	Jaccard ladder	Scale-loss hotspot similarity
E5	Adaptive + ablation	Cell-count comparison vs uniform fine
E6	Rainfall scenarios	Within-cell `PFI_h` response across intensities
E7	Sandy negative control	Coastal overlap checks (not training)

Reproducible tables and figures are released with the public repository; process detail lives in the companion research report.

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6 Results

All numbers below are copied from the live Lower Manhattan open-data pilot (`n_cells=141`). They do **not** describe citywide performance.

6.1 Spatial H3-block CV (primary blocked evaluation)

Metric	Value
n_cells	141
spatial_cv_n_folds	5
spatial_cv_n_blocks	7
spatial_cv_accuracy_mean ± std	0.784 ± 0.069
spatial_cv_f1_mean	0.866
spatial_cv_r2_mean ± std	0.030 ± 0.343
spatial_cv_mae_mean	0.332
random_split_val_accuracy (diagnostic only)	0.690
positive-class prevalence (held-out)	0.801
always-positive (majority) baseline accuracy	0.808
always-positive (majority) baseline F1	0.893

Per-fold accuracy/F1: Fold0 0.755 / 0.850; Fold1 0.760 / 0.850; Fold2 0.773 / 0.872; Fold3 0.714 / 0.813; Fold4 0.917 / 0.944.

Reading. The held-out labels are highly imbalanced (80.1% positive). A trivial always-positive majority classifier would reach mean accuracy **0.808** and mean F1 **0.893** across the same folds, which **exceeds** the model's **0.784** accuracy and **0.866** F1. Spatial blocking makes the evaluation design more defensible but does **not** by itself establish discrimination; these blocked metrics are therefore reported as a fit-and-evaluate check on the protocol, **not** as classification skill. Fold4 accuracy (0.917) coincides with a small test set (`n_test=24`, two blocks) and must not be cherry-picked. Continuous-risk R^2 (0.030 ± 0.343) is weak and is reported only for completeness. Random-split accuracy (0.690) remains diagnostic only and is **not** the primary claim. A class-imbalance-aware ranking metric (ROC-AUC) is the more appropriate discrimination summary and is listed as a follow-up (待补充).

6.2 Scale-loss Jaccard ladder (open labels)

Fine resolution R10, hotspot quantile 0.9:

coarse_res	aggregation	jaccard	f1
8	mean	0.167	0.286
8	max	1.000	1.000
8	p90	1.000	1.000
9	mean	0.977	0.988
9	max	1.000	1.000
9	p90	0.977	0.988

Reading. Under mean aggregation, parent rollup at R8 yields Jaccard **0.167** with fine R10 hotspot parents, which is **consistent with** strong scale smoothing on this open-label stack. Max/p90 retain extrema by construction (Jaccard 1.0 at R8) and therefore should not be narrated as “no scale loss.” These diagnostics are **not** a reproduction of Svellingen et al.’s PFIb Jaccard ≈ 0.14 (different labels, resolutions, and hotspot definitions).

6.3 Adaptive vs fixed / uniform fine

Field	Value
score used for screening	`PFI_h`
n_fixed_coarse (R9)	141
n_adaptive_mixed	3933
n_uniform_fine (R11)	6909
adaptive_cell_count_ratio (adaptive/uniform)	0.569
n_parents_refined	79

score_quantile	0.8

Adaptive mixed grids use about **57%** as many cells as a **uniform R11** grid (3933 vs 6909) while refining 79/141 coarse parents; relative to the **fixed R9** baseline the mixed grid uses about **27.9×** more cells (3933 vs 141). This statement concerns **cell counts only**. We do **not** report wall-clock runtime, memory, or city-scale cost savings from this pilot.

6.4 Rainfall scenarios (`PFI_h`)

141 cells × scenarios {moderate 25, heavy 40, ida_like 75, extreme 100 mm/h}; rainfall provenance remains event-raster (synthetic constant hook); assembly is open-data.

Observed: mean `PFI_h` ≈ **0.802888** for every scenario; **within-cell range across scenarios = 0**.

Interpretation: the current pilot does **not** demonstrate rainfall-conditioned discrimination. The scenario loop is correct (only `rainfall_mm_h` varies); the flat response is explained by **constant training rainfall** — all 141 training cells carry `rainfall_mm_h = 75` from the synthetic constant event-raster hook, so rainfall has zero training variance and a classifier feature importance of **0.0**. The model has never observed rainfall variation, so `PFI_h` cannot respond to it. Non-zero response requires ingested observed event rainfall (multi-intensity, non-synthetic provenance) and retraining; definition of `PFI_h(c,r)` remains binding for future runs.

6.5 Sandy negative control

Open-data assembly: n_cells=141; n_coastal=31; n_pluvial=71; n_both=23; n_coastal_only=8; frac_coastal_only≈0.057; pluvial_minus_coastal_mean_score≈0.120. Coastal overlay is **not** a training label.

6.6 Expanded open-data pilot (`manhattan_expanded`)

Metric	Value
n_cells	956
spatial_cv_n_folds	5
spatial_cv_n_blocks	28
spatial_cv_accuracy_mean ± std	0.642 ± 0.148
spatial_cv_f1_mean	0.608
spatial_cv_r2_mean ± std	0.525 ± 0.112
spatial_cv_mae_mean	0.112
random_split_val_accuracy (diagnostic only)	0.667
positive-class prevalence (held-out)	0.479
always-positive (majority) baseline accuracy	0.479

always-positive (majority) baseline F1	0.648

Per-fold accuracy/F1: Fold0 0.801 / 0.832; Fold1 0.419 / 0.442; Fold2 0.759 / 0.736; Fold3 0.516 / 0.343; Fold4 0.715 / 0.689.

Reading. The expanded extent (`manhattan_expanded`, ≈0.09° × 0.12°) is a second, larger open-data pilot (956 cells over 28 blocks), still **not** citywide. Its held-out labels are near-even (47.9% positive), in contrast to the 80% prevalence of the smaller Lower Manhattan table. Under the same H3-block protocol, the model's spatial CV accuracy **0.642 ± 0.148** exceeds the always-positive majority baseline (**0.479**), and continuous-risk R^2 (**0.525 ± 0.112**) is substantially higher than the smaller pilot's near-zero value, indicating non-trivial continuous signal at this scale. Mean F1 (**0.608**) nevertheless remains below the always-positive F1 (**0.648**), so classification discrimination in F1 terms is **not** claimed; the per-fold spread (accuracy 0.419–0.801; F1 0.343–0.832) reflects folds whose held-out blocks are majority-negative (Fold1, Fold3) versus majority-positive (Fold0, Fold4). This is reported as a robustness check on the protocol, not as citywide skill.

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7 Discussion

7.1 Implications under stated boundaries

The Manhattan pilots **indicate** that an open-label H3 table can be trained and evaluated with blocked CV, that mean rollups substantially alter fine-hotspot membership (scale-loss), and that trained-score adaptive refinement reduces uniform-fine **cell count**. These results demonstrate execution of the protocol under the stated pilot design; they do **not** establish product-ready city maps or rainfall-responsive screening under the current synthetic rainfall hook. On classification discrimination the two pilots differ: the smaller table does **not** beat its majority-class baseline, while the expanded table beats it on **accuracy** (0.642 vs 0.479) but **not** on F1 (0.608 vs 0.648), so discrimination remains only partially evidenced and is not claimed as skill.

Relative to Svellingen et al. (2026), the dialogue is conceptual: they aggregate a proprietary building index (PFIb) into H3 for scalable communication; we learn on public labels with spatial holdouts and an explicit non-PFIb `PFI_h(c,r)`. Our open-label Jaccard under mean aggregation R10→R8 (0.167) should **not** be narrated as matching their PFIb Jaccard ≈ 0.14 — different labels, resolutions, and hotspot definitions. Proximity of 0.167 to 0.14 is coincidental and must not be used as validation.

7.2 Limitations

1. **Spatial extent.** Results use two Manhattan pilots — Lower Manhattan (`n=141` cells) and an expanded extent (`n=956` cells) — neither of which is citywide New York City.
2. **Label bias.** 311 and related open indicators reflect reporting and mapping processes, not complete ground truth.
3. **Hydro proxy.** Distance-to-water from NHDPlus in a tidal/shoreline setting is a proxy, not inland drainage density.

4. **Rainfall provenance.** Event rainfall remains a synthetic constant hook; gauge/radar event ingest is 待补充.
5. **Flat `PFI_h(c,r)`.** Within-cell range across rainfall scenarios is currently **0**; rainfall-conditioned discrimination is not demonstrated.
6. **Class imbalance / no demonstrated discrimination.** The smaller pilot is highly imbalanced (80% positive); its accuracy and positive-class F1 fall **below** an always-positive majority baseline (0.808 / 0.893 vs model 0.784 / 0.866). The expanded pilot is near-even (47.9% positive) and beats the majority baseline on accuracy (0.642 vs 0.479) but **not** on F1 (0.608 vs 0.648). Classification discrimination is therefore only partially evidenced and is **not** claimed as skill; a class-imbalance-aware ranking metric (ROC-AUC) should be reported once archived.
7. **Small blocked design.** The smaller pilot distributes only 7 H3 blocks over 5 folds (per-fold test 21–49, some folds a single block); the expanded pilot uses 28 blocks (per-fold test 190–193), which is more defensible but still a limited, non-citywide design.
8. **Continuous skill.** Spatial-CV R^2 is near zero in the smaller pilot (0.030) and moderate in the expanded pilot (0.525); the latter is reported as a scale-sensitive signal, not as citywide predictive skill.
9. **Held-out sensors.** FloodNet validation is unavailable under the default stub/opt-in path.

Random-split accuracy must not displace spatial CV in claims. Fold-level variance (including Fold4 on `n_test=24` in the smaller pilot) further warns against over-interpreting a single pilot run.

7.3 Completion criteria

(i) ingest observed event rainfall with non-synthetic provenance; (ii) non-flat scenarios — within-cell `PFI_h` range > 0 across rainfall intensities on the same static \(X_c\); (iii) citywide (or larger) profile with the same spatial CV protocol — the expanded `manhattan_expanded` pilot (`n=956`) is now archived, but a full citywide run remains 待补充; (iv) FloodNet held-out validation when a non-empty layer exists.

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8 Conclusions

We present a reproducible open-label H3+ML **protocol** — spatial block CV, scale-loss diagnostics, adaptive cell-count refinement, and an explicit non-PFIb `PFI_h(c,r)` definition — evaluated on two Manhattan open-data pilots. Live metrics demonstrate execution of the protocol under the stated boundaries; they do not establish citywide operational skill. The smaller pilot does not beat a majority-class baseline; the expanded pilot beats it on accuracy but not F1 and shows moderate continuous R^2, so classification discrimination remains only partially evidenced. Observed event rainfall, non-degenerate rainfall response, a full citywide extent, class-imbalance-aware discrimination metrics, and FloodNet validation remain 待补充.

Figure captions

Figure 1. Open-label H3 pluvial flood learning protocol (workflow). Conceptual pipeline: (1) open multi-source inputs — open flood labels (DEP stormwater polygons, 311 flooding points, USGS Ida high-water marks), static predictors (elevation, slope, impervious fraction, building density, distance-to-water), a rainfall condition `r` (a synthetic constant hook in this pilot, not radar), and a FEMA Sandy negative control that is never a training label; (2) H3 assembly at R9 with provenance tags (`assembly_mode`, `feature_source`, `label_source`, `rainfall_source`); (3) learning and blocked evaluation — a gradient-boosting classifier with continuous risk regressor, H3-block `GroupKFold` spatial cross-validation as the primary blocked evaluation, a disclosed majority-class baseline, and logistic/ponding-rule baselines; (4) diagnostics and outputs — the rainfall-conditioned `PFI_h(c,r)` index, a scale-loss Jaccard ladder (R10→R9/R8), adaptive refinement screened by trained `PFI_h` (→R11), and a Sandy negative-control check.

Figure 2. Spatial H3-block cross-validation fold metrics. Per-fold classification accuracy and F1 on the Lower Manhattan pilot (`n=141` cells; 5 folds; 7 blocks). Primary reporting uses the mean ± standard deviation across folds; individual high folds (e.g. Fold4) are not interpreted as citywide skill, and accuracy/F1 are reported alongside a majority-class baseline (see Table in §6.1).

Figure 3. Open-label scale-loss across H3 resolutions. Jaccard similarity and F1 between fine-resolution (R10) hotspot parent sets and coarse (R8/R9) hotspot sets under mean, max, and p90 aggregation. Mean aggregation at R8 yields Jaccard ≈ 0.167, consistent with smoothing of local extremes; max/p90 preserve extrema by construction. Values are diagnostics on the Lower Manhattan open-label pilot and must not be equated to Svellingen et al.'s PFIb Jaccard ≈ 0.14.

Figure 4. Adaptive refinement versus uniform fine grids (cell counts). Fixed coarse (R9), adaptive mixed, and uniform fine (R11) cell counts for the pilot table (adaptive/uniform ≈ 0.569; 79 of 141 parents refined). The comparison is restricted to cell counts; it is not a wall-clock or citywide runtime claim.

Data and code availability

Public repository (code, configs, tests, paper docs, and small summary tables; large rasters/geojson, trained model binaries, and large parquet files excluded): https://github.com/Coucou2016/pluvial-flood-risk-DGGS-H3 (`pluvial-flood-risk-dggs-h3` v0.1.0). Layer provenance and download manifests are released with the repository. Synthetic demonstrations are excluded from scientific evidence. A deep process-oriented research report accompanies the manuscript in the same documentation folder.

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Additional venue-specific references 待补充.